

Felipe S. Oliveira

SOFTWARE ENGINEER · SECURITY ANALYST

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Education

State University of Campinas

Campinas, Brazil

HON. B.Sc. IN COMPUTER SCIENCE - Jan. 2011 - Dec. 2016

University of Toronto

Toronto, Canada

VISITING STUDENT - Jan. 2014 - Jan. 2015

Experience

SiDi (Samsung Institute for the Development of Informatics)

Campinas, Brazil

SYSTEMS SOFTWARE ENGINEER - May 2019 - Present

- **HDM Detector** - A new feature for the Hypervisor Device Management (HDM) Framework, part of Samsung's Knox B2B Framework. This feature tries to ensure that a given installed application in a managed device has the correct version. If not, then it takes action ranging from alerting the user or admin to blocking device features, such as the camera or wireless connections. I am responsible for researching the feasibility of the feature and implementing the first proof-of-concept. For this we add code to all layers of the Android Framework: higher level front facing Java APIs, native framework and trusted application connector in C++ and trusted application running in the ARM's TrustZone in C.
- **Bootloader Forensics** - A new security feature for the Samsung Bootloader (sbl) for the purpose of validating and detect possible modifications of key aspects of the system. This includes validation of the bootloader and overall system states, collecting relevant data for storage and encrypting and signing the information to prevent tampering. This feature works as an extension of the Verified Boot and other security features already present in the sbl.
- **Universal Credential Manager (UCM)** - Embedded Secure Element (eSE) based solution for storing credentials and using cryptographic operations over that environment. We collaborate with a team in South Korea for the extension of UCM. I act as the technical lead for this project and my team is responsible for adding new features to it's trusted application and framework, as well as doing a full refactor of the framework code to enable better testing and further extension.
- **Digital Key** - Solution for using a smartphone as a car key, implementing the specification for Digital Key issued by the Car Connectivity Consortium (CCC). We implemented a solution in ARM TrustZone for enabling secure communication between the Android Framework and the embedded secure element (eSE). We also designed and implemented many features for the Digital Key Framework, which will be available for third party developers as part of Samsung Knox.
- **Miscellaneous** - I also provide support for my team in tasks related to the implementation of trusted applications, low-level programming and systems and provide support on general security concepts.

SOFTWARE SECURITY ANALYST - May 2017 - Apr 2019

- **Bixby** - I was responsible for doing a full security assessment of Samsung's voice assistant, Bixby. This meant touching most aspects of the solution, inspecting the phone clients, servers, and researching ways to assess the security of its machine learning capabilities. This would be through novel methods, such as Adversarial Attacks on the ML models. I also conducted the development of tools to reverse engineer the gRPC protocol used for communication between the client and servers.
- **Android System Services** - Our team was responsible for the continuous security assessment of Android system services of every Samsung flagship device. We would look for vulnerabilities in these services that could enable attacks such as privilege escalation, arbitrary read/write or unauthorized access to certain features. This included writing exploits for vulnerabilities we found and writing reports that would be sent to the teams responsible for the assessed service.
- **Tizen Services and Applications** - We provided security analysis for Tizen applications and services. Tizen is a Linux-based operating system that powers Samsung wearables, smart TVs, IoT devices, among others. I was part of a small subset of our team that specialized and led security assessments in these devices.
- **Web and Android Applications and Services** - We also provided security analysis for web applications, such as Samsung Account, and Android applications, such as SHealth.

Kryptus - Solutions in Information Security

Campinas, Brazil

SOFTWARE ENGINEER - Sept. 2016 - May 2017

- **Custom Bootloader for Disk Encryption/Decryption** - We developed a full disk encryption solution based on VeraCrypt for ATMs. Unlike most solutions, in this case the key would be stored alongside the BIOS in a section of the NVRAM. We implemented custom code based on VeraCrypt's bootloader and its Windows drivers and on iPXE for remote booting and monitoring of the solution. Most of my contributions were on the bootloader and pre-boot execution environment, where we wrote baremetal code in C and x86 Assembly, but I also touched some of the work done in the drivers which are mostly written in C++. We also implemented a solution for remote management and monitoring of the ATMs in Python using the Django Framework on the server.

Extracurricular Activities and Projects

Voronator

[GITHUB.COM/FESOLIVEIRA014/VORONATOR-RS](https://github.com/FESOLIVEIRA014/VORONATOR-RS) - 2020 - Present

- Rust crate (library) implementing Voronoi Diagram and Delaunay Triangulation algorithms. It takes in a list of 2D points and returns a graph data structure containing geometrical and spatial information of either just the triangulation or both triangulation and it's dual. It also implements a variant of the Voronoi Diagram that uses centroids instead of circumcenters, allowing for a more aesthetically pleasing result with some trade-offs.

Toy Rendering Engine

[GITHUB.COM/FESOLIVEIRA014/YANVE](https://github.com/FESOLIVEIRA014/YANVE) - 2014 - Present

- Since 2014 I have been writing toy rendering and voxel engines for fun. This is the fourth iteration of the engine, providing a wrap over OpenGL, a scene graph and other useful data structures and algorithms for rendering scenes. Past iterations focused almost completely on voxel rendering and experiments on efficient meshing. However, I wanted to make a new version that enabled me to render other things without needing to re-write too much code all the time. It is written in C++17 and using *dear ImGui* for the user interface. The current version does not implement many graphics techniques yet, but past iterations (i.e. playground) implemented techniques such as *Phong Shading* and *Shadow Mapping*. Much inspiration for the design was taken from existing projects, such as magnum and Horde3D.

Honors & Awards

2016	Graduated with Distinction , Unicamp	<i>Campinas, Brazil</i>
2014	Grand Prize Winner , SPORTSHACK 2014	<i>Toronto, CA</i>
2014	Science Without Borders Scholarship , Brazilian Government	<i>Brazil</i>

Skills

Programming	C, C++, Python, Assembly (x86, ARM), Java, Rust, Javascript
Operating Systems	Android, GNU/Linux, Tizen, Windows
Disciplines	Algorithms, Data Structures, Cryptography, Security, Computer Graphics
Technologies	Git, JIRA, Perforce, Gradle, ADB, Android Studio, Visual Studio, Visual Code
Languages	English (fluent), Portuguese (native), Spanish (basic)